

Autonomous market intelligence.

Business plan.

A self-improving, evidence-gated trading engine for crypto futures and on-chain markets — engineered for net profit after every cost, audited by machine and by external review.

7

LIVE DAEMONS

17

FUSION KEYS

26k+

SEALED EVALS

2

MARKETS

01 Executive summary

Sentinel Tactical Radar is a production autonomous trading system for cryptocurrency markets. It scans a live multi-venue universe, scores candidates through a statistically-fused signal stack, sizes positions on measured evidence, protects every position with a three-daemon defense mesh, and journals every decision for audit.

The honest position. The engineering is ahead of the evidence. Sentinel's machinery — evidence-gated scoring, regime-conditional strategy weighting, empirical-Kelly sizing, inventory-aware exposure control, and a self-custody on-chain execution lane — is institutional-grade in design. Live realized performance to date is net negative (−\$54.80 all-time across 232 closes at the time of writing), which this plan reports plainly rather than disguising. The business case rests on a deliberate sequence: **prove the edge on a small live book → scale only what survives measurement → productize the infrastructure.**

Operating principle: profit is the only metric. Never win-rate alone, never gross before net. Every gate, every grade, every roadmap item is subordinate to net dollars after fees, slippage and funding.

What exists today

- **Futures lane** — Bitget USDT-M perpetuals, live since Oct 2026: 20-signal scan board, 17-key statistical fusion, longs-only mandate, bounded inventory skew, Kelly-negative vetoes.
- **On-chain lane** — self-custody Solana wallet trading DEX candidates via Jupiter (gasless-capable), with Rugcheck screening, purse caps, and software trails. Every fill is a solscan-verifiable signature.
- **Protection mesh** — three daemons: execution supervisor, 10s protection watcher (band-clamped stop repair, staggered TP ladders), and a ~5s liquidation guard (crash brake, liq-zone trims, naked-stop synthesis).
- **Wealth vault** — 25% carry on net-new equity highs, high-water-mark gated, physically segregated from the tradable book.
- **Attribution layer** — "Pimp City": every fill assigned to a persona, per-pimp and per-girl P&L, vault-carry KPIs, form/mood evolution — the ledger that answers "who made the money."
- **Ops surface** — live dashboard, Telegram command-and-control, VIP signal ingestion as confluence, TradingView webhooks, integrity checksums on all state.

02 The system

SCAN — ~200 instruments/cycle (CEX perps + DEX pairs across Solana/Base/BSC). Regime classification, microprice + queue imbalance, BTC-relative strength, funding/carry, social + VIP confluence, strategy-library edge priors (390k external backtests indexed per symbol).

SCORE — 17-key fusion: serially-demeaned components, cross-sectional z-scores, Bayesian-ridge residual weighting onto realized alpha (shrunk to priors, $n/(n+300)$ evidence gate).

SIZE — empirical Kelly per strategy ($f^* = p - q/b$, half-Kelly deployed), inventory skew $\exp(-\gamma \cdot q_{\text{eff}})$ with $1.5\times$ correlated-name weight, hard risk ceilings independent of conviction.

PROTECT — exchange-side stops + software mesh: crash brake ($\geq 0.8\%/5s$ gap), liq-zone trims, fee-lock ratchet, TP ladders, stall-cuts, margin-loss cap, manual-hold arbitration.

Signal-to-execution pipeline

STAGE	MECHANISM	HONEST STATUS
Discovery	Multi-venue universe scan, exec-catalog gating	Live, verified
Evidence	26,344 forward-labeled signal evals; grade + outcome joins	Large n, autocorrelated — independence clustering queued
Fusion	17-key ridge, synthetic test $R^2=0.98$	Math verified; live predictive alpha unproven (IC24h ≈ 0.005)
Gates	RR floors, drift guard, meta-label p(profit), family/kelly vetoes, deny lists	Live; rejects journaled with named gates
Execution	Stop/plan-triggered entries, leverage from stop width, markout grading	Live
Exit	Fee-aware trails, TP ladders, ratchets — tighten-only	Live, net-RR enforced
Accounting	Per-fill journal, episode joins, per-strategy/per-persona attribution	Live; fee % disclosed

External review built in. An independent critical review (ChatGPT-graded, via the project's external advisor) is treated as a defect tracker: its October review produced a live code change — a measured-negative Kelly cell now vetoes outright rather than taking a $0.3\times$ probe size.

03 Edge machinery

Sentinel's stack is built around one desk rule: **never widen the proof bar to widen the funnel**. Components compete on measured marginal contribution, and measured-negative cells lose their bullets.

The 17-key fusion

- **Serial demeaning** — each component minus its ~30-cycle EMA; only the surprise carries signal.
- **Cross-sectional z** — every key z-scored across the board each cycle (± 4 clamp); ranks over raw units.
- **Residual weighting** — Bayesian ridge of the z-map onto realized 24h alpha, $\lambda=30$ shrunk to hand priors. The momentum/volume/surge triple-count dies here — each key earns only its marginal lift.
- **Evidence ramp** — blend = $n/(n+300)$, capped 0.7. Hand-tuned prior is the floor until sealed evals accumulate.

Standout mechanisms

MECHANISM	FUNCTION	SOURCE BASIS
BTC-pair confluence	± 4 tilt: up vs USDT but down vs BTC = pure beta, penalized	Relative-strength doctrine
Regime strategy weights	Strategy $\times$ tape-type cross-tab; regime-local overrides once cells mature	Post-regime audit
Inventory skew	$\exp(-2.2 \cdot q_{\text{eff}})$ size decay vs same-direction deployed margin	Avellaneda–Stoikov (Stanford MS&E 448)
Microprice + OBI	Top-5 book: micro-vs-mid in half-spread units + queue imbalance	Microstructure lit
Library edge priors	390k-strategy public leaderboard density per symbol $\rightarrow \pm 3.5$ prior	trader.dev public API
Kelly vetoes	Per-strategy $f^* \leq 0 \rightarrow$ full stand-down (post-review hardening)	Empirical Kelly

Ablation commitment. The next major build is a shadow reference: every signal scored by the frozen v1.0-master logic alongside the current stack on identical tape, both forward-marked — a controlled A/B so improvements are measured, not asserted.

04 Current performance — reported plainly

This section reports the live journal exactly as the ledger stores it. Nothing here is projected.

-\$54.80 NET REALIZED, ALL-TIME	232 CLOSES JOURNALED	0.22 PROFIT FACTOR
\$23.85 TOTAL FEES PAID	23.7% WIN RATE	~100% MAX DRAWDOWN OF FUNDED EQUITY

Attribution

BOOK	CLOSES	NET USD	FEES	PF
Bot-attributed	48	-\$4.18	\$0.91	0.16
Manual / foreign fills	184	-\$50.62	\$22.94	0.23

The journal separates machine decisions from operator interventions. The largest loss clusters were correlated manual campaigns (e.g. concentrated same-direction alt exposure), which is precisely the failure class the new inventory-skew and corr-cluster machinery now gates.

What the eval engine says about its own score

METRIC (N=26,344 EVALS)	VALUE	READING
Directional accuracy, 24h	52.6%	marginally better than chance
Information coefficient, 24h	0.005	≈ zero predictive alpha in raw score
A-grade cohort hit rate	25.2%	inverted vs B-grades (37.7%)
Avg alpha vs BTC, 24h	-0.04%	slightly negative pre-fusion

The thesis being tested: the fusion layer exists because the hand score demonstrably lacks predictive alpha. If the ridge + evidence stack cannot produce a measurable lift in IC and net \$/trade over the next evidence window, the correct engineering outcome is to say so — and this plan budgets for that honest answer.

05 Market & business model

The opportunity

Retail crypto trading remains dominated by discretionary losses and unsystematic bots. The durable gap is not signal generation — it is **audited, cost-aware, self-evaluating execution**. Sentinel's moat, if it earns one, is the evidence pipeline: every signal sealed and labeled, every strategy scored on its own realized record, every fee accounted.

Phased model

PHASE	BUSINESS	TRIGGER TO ENTER
0 · Now	Proprietary micro-book. Small live capital, maximal instrumentation.	— in progress
1 · Proof	Scaled prop trading. Vault carry compounds internally.	3+ months positive net PF>1.15 after costs, IC lift demonstrated
2 · Signals	Subscription signal board (the scan stack already produces ranked, graded outputs with sealed-eval audit trails).	Demonstrated eval outperformance vs grade baseline
3 · Platform	Managed vault / copy-trading surface; on-chain lane productized as audited micro-cap discovery.	Regulatory review + sustained Phase-1 record

Unit economics (targets, not history)

<15% FEES AS % OF GROSS P&L TARGET	>\$0.15 NET PER CLOSE TARGET	>1.15 PROFIT FACTOR GATE TO SCALE
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Cost structure

- VPS hosting ~\$15/mo · API keys \$0 · data feeds \$0 (public APIs)
- Trading fees: ~0.06%/side futures taker; DEX swaps 0.15–0.4% + spread
- Dominant cost center to date: fee drag on high-churn books — engineered down via net-RR floors and churn breakers

06 Risk — the pre-mortem is written

Sentinel keeps a standing pre-mortem register: every way the system can lose money, mapped to its guard. Highlights:

FAILURE MODE	GUARD
Adverse selection on entries	Stop/plan triggers into the level, RR re-check at fill, markout grading
Redeploy-after-loss churn	Entry dedup, auto-quarantine (4 losers + \$1 bleed → 24h deny), account breakers
Correlated stack = one bet	Corr-cluster veto + continuous inventory skew (1.5× corr weight)
Cascade faster than polling	Crash brake + exchange-side stops (VPS outage never leaves the book naked)
Profit round-tripping	Fee-lock ratchet, dense trail tiers, vanish-close on giveback
Fee churn	Dust-floor clip skips, fee-burn breaker, net-RR floors include costs
Edge evaporation	CUSUM edge-death + Kelly vetoes on measured-negative cells
Stale feed as truth	WS marks anchored to REST, >5% deviation rejected
On-chain tail risk	Purse caps, Rugcheck screen, audit pass, software trails (disclosed limit: no exchange-side stop exists on-chain)
Operator override decay	Bounded self-expiring override channel

Known limitations — disclosed, not hidden

- API key is futures-trade-only: vault transfers to spot require manual action or a perms upgrade (documented).
- Software on-chain trails can fail when liquidity evaporates; Rugcheck is a screen, not a guarantee.
- Signal evals are hypothetical forward-marks, not fill-linked; independence clustering for eval evidence is queued.
- No ablation harness yet — frozen reference exists, live shadow scoring is the next build.

07 Roadmap & governance

Now → next evidence window

ITEM	TYPE	STATUS
Shadow A/B scorer (v1.0 vs current, identical tape)	Evidence	next build
Eval independence clustering (symbol+day buckets)	Evidence	queued
Fee/slippage decomposition in eval labels	Accounting	queued
On-chain lane first fills under audit	Live	armed, purse funded
Kelly-negative veto + inventory skew	Risk	deployed
Vault physical transfer	Ops	awaiting API perms / manual

Governance

- **Integrity manifest** — CRC checksums on all mutable state; tamper-evident journals.
- **Every fill verifiable** — exchange fills journaled; on-chain fills carry solscan signatures.
- **External review loop** — independent critiques (W. Evans) triaged to code changes, evidenced by commit hash.
- **Standing mandates** — longs-only, min-leverage band, profit-first grading, "cut losers quickly" enforcement, take-profit discipline.

The operating thesis

"You've reached a point where demonstrating that Sentinel works is more important than adding further capabilities."

— external review, 10 Oct 2026 (adopted as the standing order)

The next report is a measured-live-evidence report — or an honest account of why the edge isn't there. Either way, the journal decides.